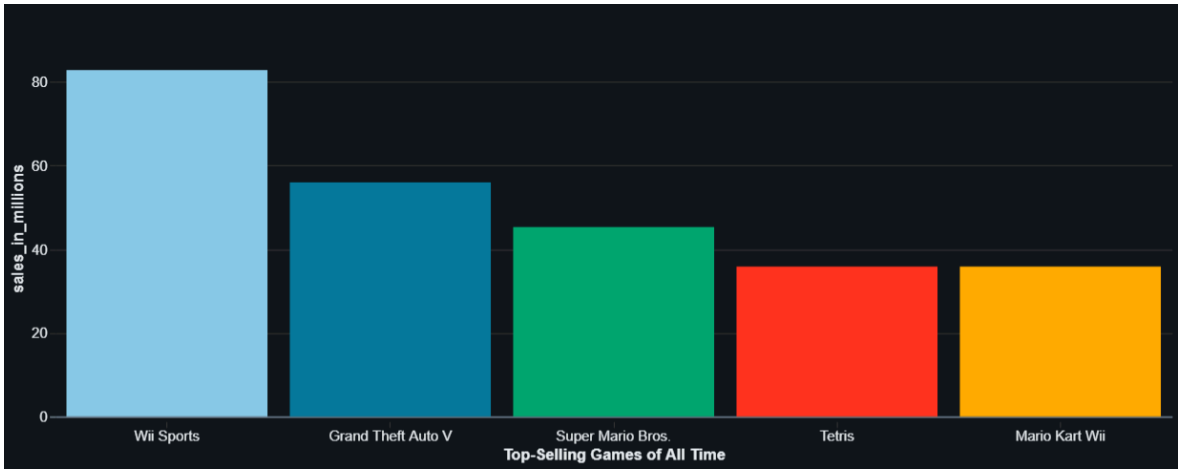
Video Game SQL Project:
Dataset Overview and Schema Documentation

Introduction

This dataset analyzes global video game sales from 1980 to 2020. It was sourced from *Kaggle.com* and compiled by Anand Shaw. The original CSV file, *Video Game Sales*, is publicly available for download on Kaggle.

The video game industry has become one of the most influential forces in global entertainment, shaping pop culture across generations. Iconic characters such as **Mario**, **Sonic the Hedgehog**, and **Link** from *The Legend of Zelda* have transcended gaming to appear in television, film, merchandise, and other media worldwide.



Despite its cultural impact, the industry remains highly competitive. Most publishers active in 1980 no longer exist today. A single hit title can define a company's success one year and lead to losses the next. High overhead costs, specialized labor, and rapid technological change make sustainability a constant challenge.

This analysis aims to provide actionable insights for emerging developers seeking market entry and for established companies like **Nintendo Inc.** looking to strengthen their global position. All SQL queries, data cleaning, and visualizations were conducted using **Databricks**.

Explanation of Dataset and its Values

The *Vgsales* dataset ("the Dataset") was downloaded from *Kaggle* as a Microsoft Excel Comma-Separated Values (CSV) file and uploaded to a *Databricks* workspace for cleaning and analysis. It records global video game sales across four major regions — North America,

Europe, Japan, and Other — spanning the years 1980 to 2020. Each entry includes the game's title, publisher, genre, and regional sales figures, providing a comprehensive view of industry performance over four decades.

Columns

- **Rank:** Number assigned to rank video game titles by sales figures. Example: the top-selling game is ranked 1, the second-highest is ranked 2. *Data type:* `BIGINT`
- **Name:** Title of the video game. *Data type:* `STRING`
- **Platform:** Console or system on which the game was released. *Data type:* `STRING`
- **Year:** Year the game was first released. Stored as `STRING`; may be cast to `INT` during cleaning.
- **Genre:** Category or type of game (e.g., Action, Sports, Shooter). *Data type:* `STRING`
- **Publisher:** Company responsible for publishing the game. *Data type:* `STRING`
- **NA_Sales:** Units sold in North America (in millions). *Data type:* `DOUBLE`
- **EU_Sales:** Units sold in Europe (in millions). *Data type:* `DOUBLE`
- **JP_Sales:** Units sold in Japan (in millions). *Data type:* `DOUBLE`
- **Other_Sales:** Units sold in regions not listed above (in millions). *Data type:* `DOUBLE`
- **Global_Sales:** Total worldwide units sold (in millions). Calculated as the sum of all regional sales. *Data type:* `DOUBLE`

Data Cleaning and Validation

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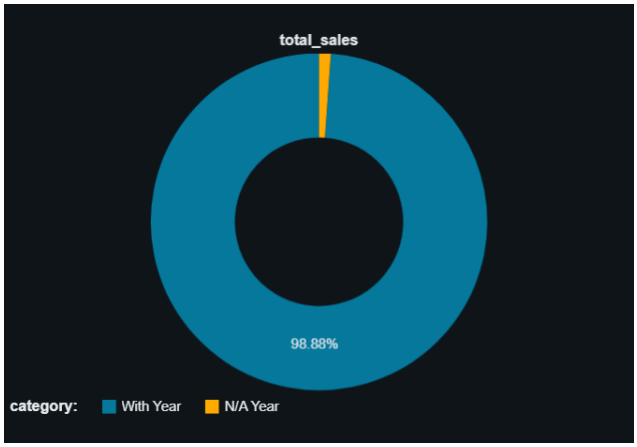
Before analysis, the dataset underwent thorough cleaning and validation to ensure accuracy and reliability. These checks help guarantee high-quality insights for end users.

- **Rank:** No duplicate or null values were found, as expected.
- **Name:** A potential data issue was identified — many titles appear across multiple platforms, resulting in duplicate names in separate records. This overlap may affect total sales calculations, both regionally and globally.
- **Platform:** The column contains 31 distinct values. All platform names are abbreviated and may require revision for clarity. For example, "XB" should be changed to "Xbox," and "GB" to "Game Boy."
- **Year:** The column includes 271 `N/A` values. Although manual correction using `UPDATE` and `SET` commands was considered, replacing these values would have minimal impact on overall analysis.
- **Impact of Missing Years:** Titles with `N/A` years account for only **1.13%** of global sales between 1980 and 2020. Given the low proportion, correcting these entries would not justify the time and resources required.

Table [redacted]

	1.2 round(na_sales, 2)	1.2 round(sales_with_years, 2)	1.2 proportion_pct
1	100.08	8820.36	1.13



The *Year* column is currently stored as a `STRING` but should be classified as an `INTEGER`. Existing `N/A` values prevent the `ALTER COLUMN` command from executing successfully. To address this, the `try_cast()` function will be used to safely convert valid year entries to `INT` while filtering out `N/A` values during analysis.

- **Genre:** Contains 12 distinct video game categories. No data quality issues detected.
- **Publisher:** Includes **203 "Unknown"** values and **58 "N/A"** entries. After review, these were left unchanged, as replacing them would be time-consuming and offer minimal analytical benefit. The focus remains on top-selling publishers with valid entries.
- **Publisher Name Variations:** Minor inconsistencies found — for example, "Milestone s.r.l." appears as both "Milestone s.r.l" and "Milestone." These variations were retained since total sales, rather than publisher name uniformity, are central to this analysis.
- **NA_Sales:** No quality issues detected. Column will be aliased as `sales_in_millions` for clarity.
- **EU_Sales:** No quality issues detected. Column will be aliased as `sales_in_millions` for clarity.
- **JP_Sales:** No quality issues detected. Column will be aliased as `sales_in_millions` for clarity.
- **Other_Sales:** No quality issues detected. Column will be aliased as `sales_in_millions` for clarity.
- **Global_Sales:** No quality issues detected. Column will be aliased as `sales_in_millions` for clarity.

Key Insights and Analysis

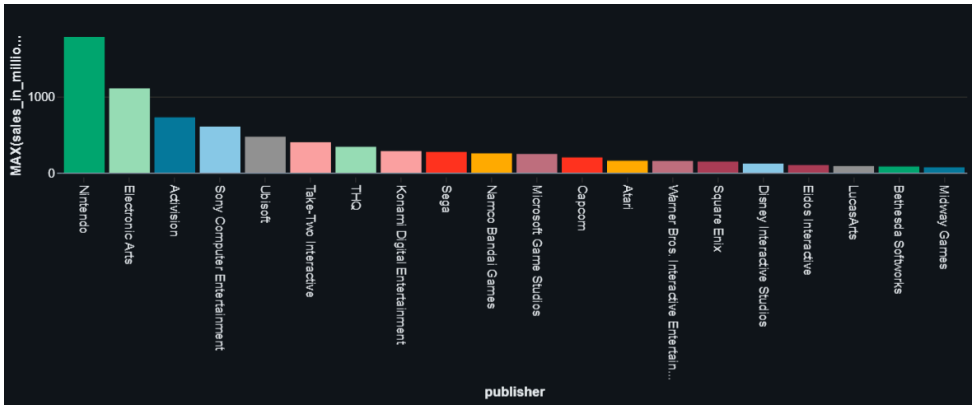
After conducting SQL-based analysis and applying various functions, several notable insights emerged from the dataset:

1. The video game industry remains **highly competitive** and volatile.

- Of **579 publishers** active between 1980 and 2020, only **two** — *Atari* and *Activision* — maintained continuous operations throughout the entire period.
- The **Top 3 publishers** — *Nintendo*, *Electronic Arts*, and *Activision* — collectively account for **approximately 40%** of global video game sales from 1980 to 2020.
- High overhead costs, skilled labor requirements, and aggressive marketing demands contribute to the industry's challenging business environment.
- New developers face significant barriers to entry due to competition from established conglomerates and shifting consumer preferences.

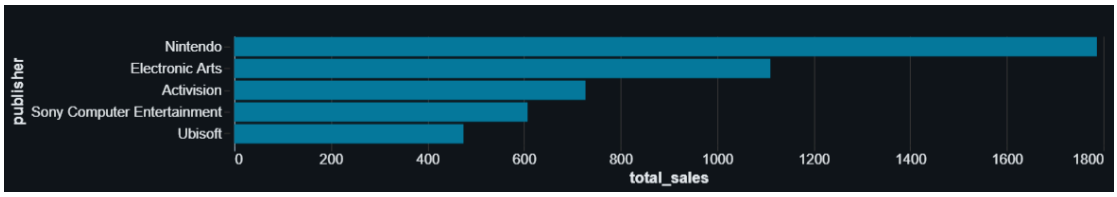
Table

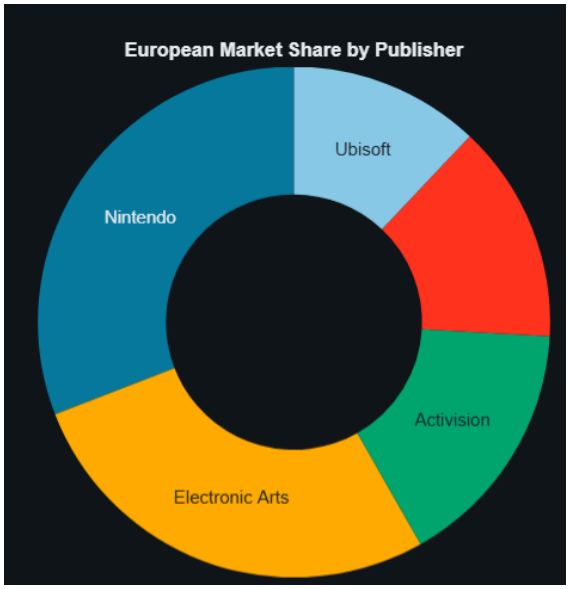
	publisher	publisher_sales	percentage_of_total
1	Total (3 Publishers)	3624.34	40.63
2	Nintendo	1786.56	20.03
3	Electronic Arts	1110.32	12.45
4	Activision	727.46	8.15

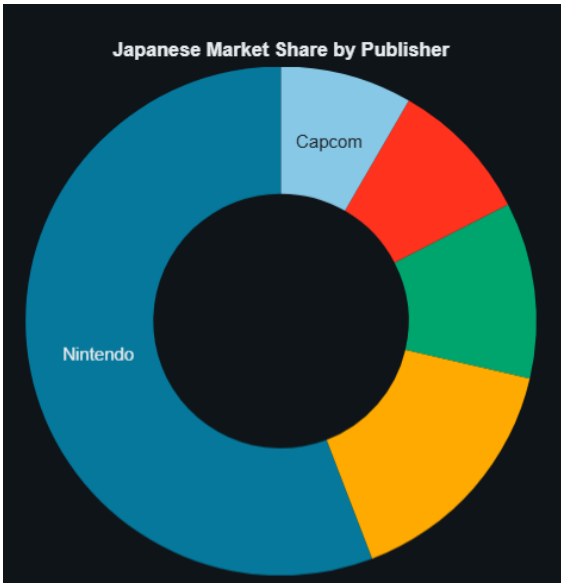


1. **Nintendo largely dominates the North American, European and Japanese markets.**

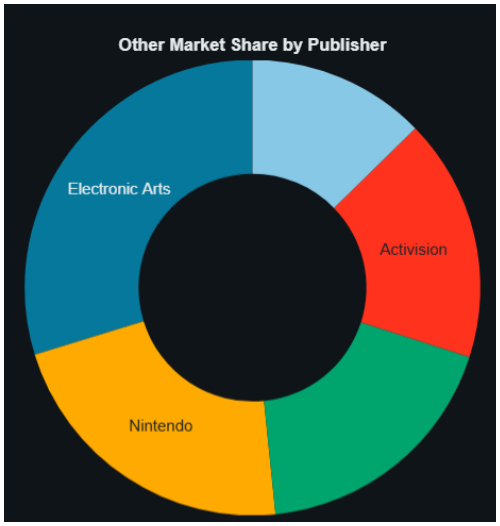
Due to brand recognition and iconic IPs, Nintendo has enjoyed longstanding success in the three main markets of the global video game trade.







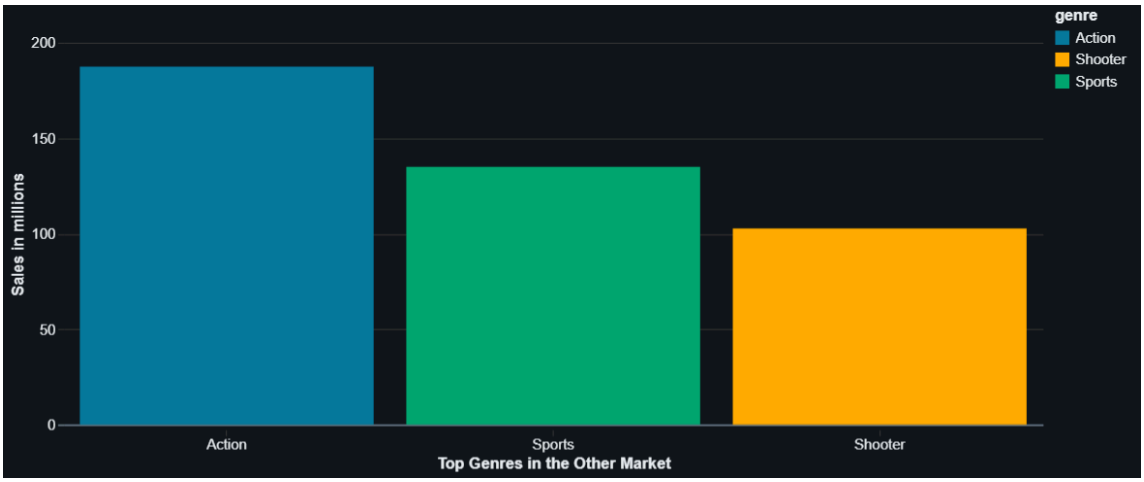
However, Electronic Arts dominates Nintendo in the regions outside of North America, Europe and Japan (the “Other” market).

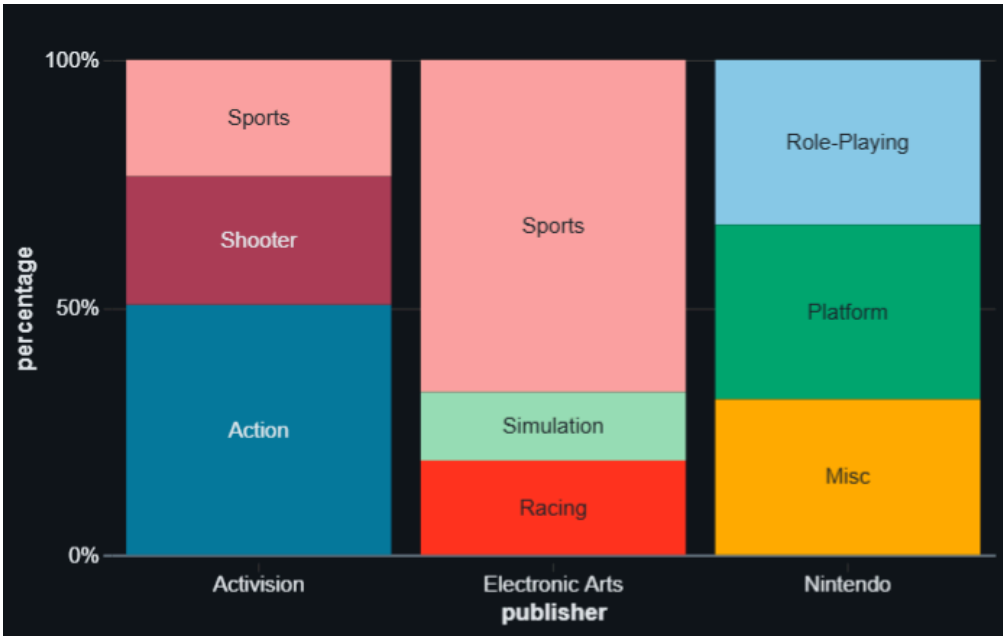


2. **Nintendo, while dominant, struggles in key genres.**

   While Nintendo remains a dominant force in global video game sales, it continues to face challenges in key genres. The most popular categories across North America, Europe, and other territories are **Action**, **Sports**, and **Shooter** — areas where Nintendo's presence is comparatively limited. The company's focus on family-friendly and role-playing titles has restricted its reach in these high-demand segments.

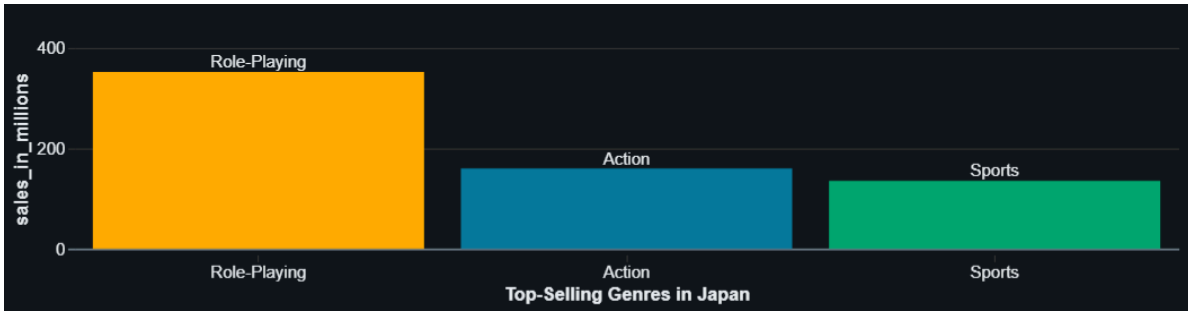
   Nintendo's best-selling game, *Wii Sports*, was a global success but represented a departure from its traditional RPG-oriented catalog. Expanding into stylized **Shooter** and **Sports** titles could help Nintendo strengthen its position in markets where these genres dominate.

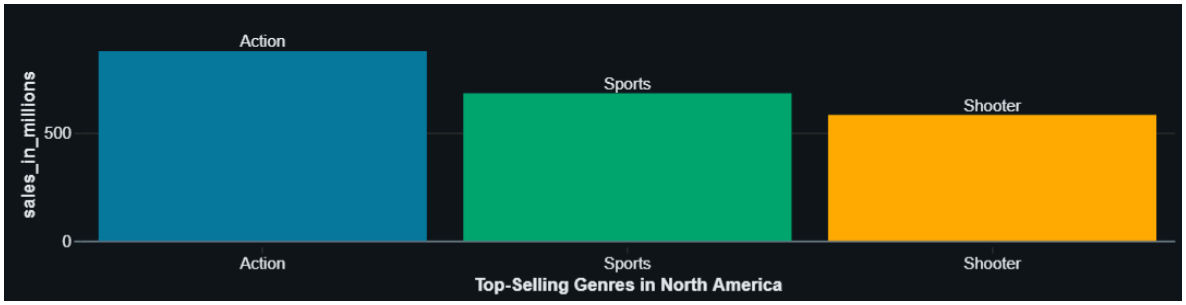




3. Japan's game preferences differ from the rest of the world

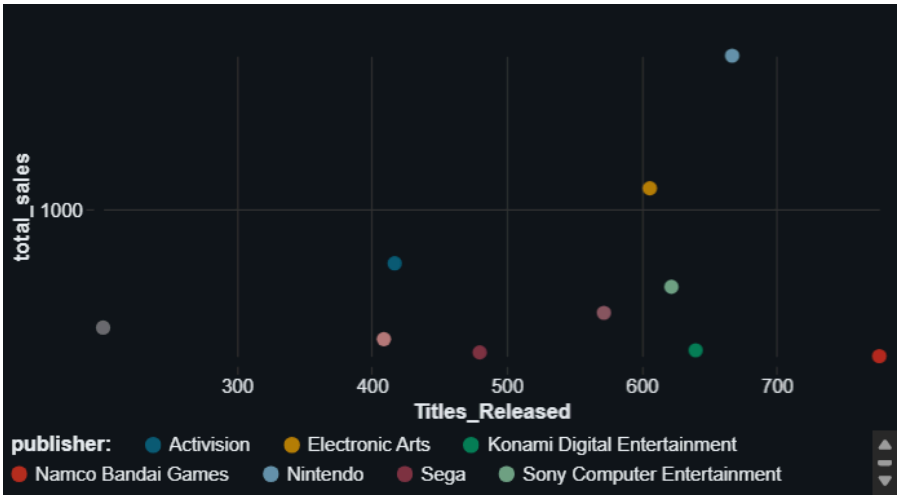
Japan's gaming market shows distinct genre preferences compared to global trends. The **Role-Playing Game (RPG)** genre dominates Japanese sales, reflecting strong cultural affinity for narrative-driven gameplay. In contrast, **Shooter** titles — which rank among the top genres worldwide — are only **#12** in Japan. This disparity helps explain Nintendo's continued emphasis on RPG development and its limited focus on Shooter titles.





4. **There is no correlation between quantity of titles and total sales.**

   There is **no strong correlation** between the number of game titles released and total sales. The scatterplot indicates only a **loose relationship**, suggesting that producing more titles does not necessarily lead to higher overall revenue. This finding implies that the market values **quality and brand strength** over sheer quantity of releases.



Business Recommendations

1. **For Nintendo, they need to invest in the Shooter genre to cement their market dominance worldwide.**

   - **Expand into the Shooter genre:** To strengthen global market dominance, Nintendo should invest in developing stylized **Shooter** titles. Establishing a dedicated studio focused on this genre would allow innovation while maintaining brand identity.
   - **Balance innovation with brand image:** Cultural and traditional norms in Japan emphasize Nintendo's "family-friendly" reputation. The company avoids mature content typical of franchises like *Call of Duty* to preserve this image.

- **Leverage existing success:** Nintendo's *Splatoon* (2015) demonstrated that Shooter games can align with its values by replacing lethal weaponry with playful mechanics such as water guns. Building on this model could expand appeal without compromising reputation.
- **Penetrate the Sports market:** Increasing production of **Sports** titles would diversify Nintendo's portfolio and capture audiences in regions where Sports games dominate sales.

2. **For new developers, focus on the Other market, build up your reputation and go global.**

    - **Focus on emerging markets:** New developers should prioritize the *Other* regions — including South America, the Middle East, and parts of Asia — where consumer loyalty to major publishers like Nintendo is lower and players are more open to exploring new titles.
    - **Build reputation locally, then scale globally:** Establishing a strong regional presence can serve as a launchpad for global expansion.
    - **Invest in localization:** Spanish localization is essential for reaching *hispanophone* gamers in Latin America and Spain.
    - **Understand cultural preferences:** Conducting research into regional gaming habits and cultural norms will help tailor content and marketing strategies to resonate with diverse audiences.

3. **For new RPG developers, focus on Japan.**
4.

- **Target the Japanese market:** New developers, particularly those based in North America, should prioritize Japan when launching Role-Playing Games (RPGs). The genre's strong cultural resonance and sustained popularity make Japan the most promising market for new RPG titles.
- **Avoid over-focusing on Western regions:** Marketing exclusively in North America or Europe yields limited returns for RPGs, as these regions favor Action, Sports, and Shooter genres.
- **Invest in cultural expertise:** Developers should hire **Japanese culture consultants** and **localization specialists** to adapt storylines, dialogue, and marketing materials for Japanese audiences. Authentic cultural representation and language accuracy will significantly improve market reception and brand credibility.

Conclusion

SQL analysis of global video game sales reveals clear strategic opportunities for new publishers. Nintendo's dominance in family-friendly genres contrasts with its limited presence in Shooter and Sports titles, highlighting room for expansion. Data-driven insights from this project

demonstrate how structured querying and visualization can guide smarter business decisions and genre diversification in a rapidly evolving and competitive market.

Methods and SQL Code for Analysis

Finding duplicate values in Rank column

```sql
select rank, count(*) as duplicates
from workspace.default.vgsales
group by Rank
having duplicates > 1
```

Finding proportion of N/A Year sales to sales with Year values

```sql
WITH NA_Year_Sales AS (
  SELECT SUM(global_sales) AS na_sales
  FROM workspace.default.vgsales
  WHERE year = 'N/A'
),
year_sales AS (
  SELECT SUM(global_sales) AS sales_with_years
  FROM workspace.default.vgsales
  WHERE year != 'N/A'
)
SELECT
  round(na_sales, 2),
  round(sales_with_years, 2),
  ROUND(try_divide(na_sales, sales_with_years) * 100, 2) AS proportion_pct
FROM NA_Year_Sales
CROSS JOIN year_sales
```

Find Shared Proportion of Top 3 Publishers

```sql
WITH publisher_sales AS (
  SELECT
    publisher,
    SUM(global_sales) AS total_sales
  FROM workspace.default.vgsales
  WHERE publisher IN ('Nintendo', 'Activision', 'Electronic Arts')
  GROUP BY publisher
),
total_sales AS (
  SELECT SUM(global_sales) AS all_sales
  FROM workspace.default.vgsales
)
SELECT
  publisher,
  ROUND(total_sales, 2) AS publisher_sales,
  ROUND(total_sales / all_sales * 100, 2) AS percentage_of_total
FROM publisher_sales
CROSS JOIN total_sales
UNION ALL
SELECT
  'Total (3 Publishers)' AS publisher,
  ROUND(SUM(total_sales), 2) AS publisher_sales,
  ROUND(SUM(total_sales) / MAX(all_sales) * 100, 2) AS percentage_of_total
FROM publisher_sales
CROSS JOIN total_sales
ORDER BY publisher_sales DESC
```

Finding Top Three Genres Per Publisher

```sql
WITH ranked_genres AS (
  SELECT
      publisher,
      genre,
      COUNT(distinct name) AS titles_per_genre,
      ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (PARTITION BY publisher), 2) AS percentage,
      ROW_NUMBER() OVER (PARTITION BY publisher ORDER BY COUNT(distinct name) DESC) AS rank
  FROM workspace.default.vgsales
  WHERE publisher IN ('Nintendo', 'Activision', 'Electronic Arts')
  GROUP BY publisher, genre
)
SELECT
    publisher,
    genre,
    titles_per_genre,
    percentage
FROM ranked_genres
WHERE rank <= 3
ORDER BY publisher, titles_per_genre DESC;
```

References

- Shaw, A. (n.d.). *Video Game Sales Dataset.* Kaggle. Retrieved from https://www.kaggle.com/datasets/ashaheedq/video-game-sales